

Frugal probabilistic wind forecasting, farm-power estimation, and robust offshore siting

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Sea Winds Predictions, Phase 2

Abstract

This study addresses three coupled outputs: probabilistic forecasting of 125 m wind on a 1.3 km grid, selection of a legal 1.21 GW offshore wind farm, and probabilistic farm-energy estimation for bidding. The forecasting system combines compact quantile model-output statistics, vector-aware direction correction, fine-grid downscaling, conformal residual calibration, and causal GraphCast trajectory fields. Optional corrections are retained only when chronological replay reduces interval score while satisfying year, issue-time, spatial, and physical-regime gates. The submitted forecast contains all 4,196,640 required rows and was matched completely by the competition scorer. Its reported interval scores are 9.511, 17.124, and 16.835 for wind speed and 76.400, 335.571, and 315.183 for wind direction at one, seven, and fourteen days, respectively; the platform primary score is 1.497276.

The siting procedure first screens every legal centre represented in the supplied static grid and then applies exact five-year PyWake replay to a diverse shortlist. A wake-aware, constrained coordinate search refines the 55-turbine layout. The selected centre is 54.109676°N, 0.952316°E, where the supplied-grid depth is 44.14 m. Historical replay produces a mean capacity factor of 50.36%, mean annual energy production of 5,337.7 GWh, and mean wake loss of 7.10%. A like-for-like public-data proxy improves from 47.08% to 50.36% mean capacity factor and from 9.70% to 7.10% mean wake loss. For bidding, the final site-level wind quantiles are propagated through the prescribed hub-height transformation, turbine power curve, and wake model to produce 96 ordered six-hour farm-energy triplets. Training and inference are implemented as two Python entry points and use compact, single-threaded statistical estimators.

Keywords: probabilistic wind forecasting; farm-power uncertainty; model-output statistics; circular regression; offshore wind siting; wake optimization; computational frugality

1 Introduction

Offshore wind decisions require both atmospheric prediction and spatial design. Forecasts must express uncertainty at several lead times, while a wind-farm design must convert a long-term wind climate into high energy yield without violating depth, footprint, or turbine-spacing constraints. The two tasks have different failure modes. Forecast intervals can be sharp yet under-cover unusual conditions; a farm layout can perform well for the mean wind rose yet lose that advantage in an individual year or after a modest directional shift.

The competition therefore rewards more than a single point estimate. Forecasts are evaluated at one, seven, and fourteen days using interval scores for wind speed and circular wind direction. Siting is evaluated for a fixed set of 55 IEA 22 MW turbines using a hidden wind climate and the prescribed PyWake configuration, while optional farm-power quantiles connect the atmospheric forecast to bidding. This work asks four practical questions: (i) can compact statistical calibration extract transferable information from the supplied forecasts and an allowed ERA5-trained weather model; (ii) can optional forecast corrections be screened against distributional instability rather than selected by aggregate score alone; (iii) can a legal site and layout be improved while preserving performance across years and plausible wind-direction perturbations; and (iv) can wind uncertainty be converted into a physically bounded energy forecast without adding another fitted model?

The main contributions are: a lead-specific probabilistic calibration pipeline that treats direction as a circular variable; a strict promotion protocol that records both improvements and rejected branches; a joint centre-layout siting procedure supported by exact historical replay, a reproducible same-data comparator, and several robustness tests; and a physical wind-to-power coupling that uses the selected layout itself to construct the optional bidding quantiles. The emphasis throughout is on causal inputs, reproducibility, and economical computation.

2 Data and evaluation protocol

2.1 Official data and prediction domain

The fitted system uses the official high-resolution 125 m target, low-resolution reanalysis, ECMWF HRES tables, coarse wind fields, bathymetry, coastline diagnostics, and static grids supplied for the competition [1, 2, 3]. The

definitive inference predictors are taken from the separately released eight-window dataset [4]. Reanalysis provides retrospective atmospheric fields, whereas HRES provides forecast information associated with an official issue time. Historical targets are used only for model fitting and retrospective validation.

The target grid contains 479×433 cells, including 75,653 sea cells; the scored footprint contains 43,715 points. Eight inference windows, three lead times, and four forecast hours require

$$43,715 \times 8 \times 3 \times 4 = 4,196,640$$

prediction rows. Before archiving, the inference program verifies complete and unique keys, finite outputs, non-negative and ordered speed quantiles, and valid circular direction values.

2.2 Forecast score

Each speed prediction contains the lower endpoint, median, and upper endpoint of a nominal 90% interval. For an observation y , endpoints l and u , and $\alpha = 0.10$, the linear interval score is

$$S_\alpha(l, u; y) = (u - l) + \frac{2}{\alpha}(l - y)\mathbf{1}\{y < l\} + \frac{2}{\alpha}(y - u)\mathbf{1}\{y > u\}.$$

It rewards narrow intervals but penalizes observations below or above them [5, 6]. Direction uses the corresponding circular construction because 0° and 360° denote the same orientation. Lower scores are better. The platform primary score is rank based and is therefore not the arithmetic mean of the six raw interval scores.

2.3 Siting objective and constraints

The farm contains exactly 55 IEA 22 MW turbines, for a fixed rated capacity of 1,210 MW. The centre must be over sea, within the prescribed North Sea and common weather footprint, and at a supplied-grid water depth no greater than 50 m. All turbines must lie within $\pm 7,500$ m of the centre in both coordinates, and every pair must be separated by at least 1,420 m. With capacity fixed, the principal yield statistic is

$$\text{CF} = \frac{\text{AEP}}{1210 \text{ MW} \times 8760 \text{ h}}.$$

Wake loss and the official economic outputs are retained as secondary diagnostics.

2.4 Farm-energy output

The optional bidding output contains lower, median, and upper farm-energy estimates for every inference combination of window, lead time, and forecast hour. Eight windows, three lead times, and four hours give 96 ordered triplets. Each value represents energy over a six-hour forecast step and is therefore expressed in MWh. Because installed capacity is fixed at 1,210 MW, every value must lie in

$$0 \leq E_t \leq 1210 \text{ MW} \times 6 \text{ h} = 7,260 \text{ MWh}.$$

These values are generated from the final wind forecast and the submitted farm geometry; they are not independent post-processed estimates or inputs copied from an earlier submission.

2.5 Causal boundary and historical validation

Inference uses only information available at the relevant official issue time. Historical target values do not enter inference. GraphCast fields are selected from forecasts initialized at the official issue time and only at positive lead hours. The definitive reference values are never read by either executable.

Validation is chronological and component specific. Different branches have different availability and therefore different historical partitions: the core models use the official 2016–2020 target; GraphCast stress tests use held 2018 and 2020 periods; one-day context endpoints use 2019 and 2020; seven-day conditional endpoints use four folds from 2017–2020; and fourteen-day endpoint replay uses 2019 and 2020. All promoted policies are replayed on the exact 1.3 km footprint used for scoring. A branch must improve aggregate interval score and pass its recorded worst-year, issue-time, spatial, and physical-regime gates. These diagnostics are selection-conditioned because the same historical folds inform branch promotion; they are robustness checks, not unbiased estimates of future performance or formal confidence intervals.

3 Forecasting method

3.1 System architecture

Figure 1 summarizes the information flow. The design separates fitted statistical estimators from deterministic, audited deployment policies. This distinction is important for both scientific interpretation and computational accounting.

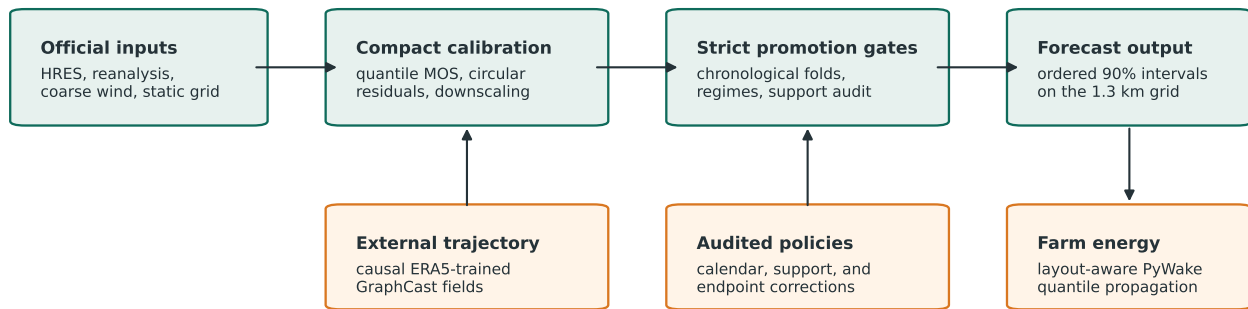


Figure 1: Integrated workflow. Official predictors and declared causal trajectory fields enter compact statistical calibration. Optional branches are deployed only after chronological and regime-based validation. The gridded wind intervals also drive a physical farm-energy transformation using the selected layout.

The artifact contains 29 fitted estimators (Table 1). This count excludes deterministic policies such as calendar-hour masks, historical-support checks, fixed endpoint transformations, branch selectors, circular blends, quantile sorting, and non-negativity clipping. Those policies are part of the deployed algorithm, but they do not require separate model fitting.

Fitted estimator family	Count	Function
Quantile model-output statistics	9	Separate speed q05, q50, and q95 estimators at three lead times.
Support-gated quantile refinements	6	One- and seven-day residual quantile corrections in supported cells.
Context endpoint models	5	Lead-specific interval correction from lagged and issue-time context.
Circular direction residual models	4	Sine and cosine residual correction without a discontinuity at north.
Shared spatial direction model	1	Compact ridge representation of geographically coherent direction error.
Conditional seven-day endpoint models	2	Lower and upper residual quantiles for stable calendar and physical regimes.
Fine-grid vector downscalers	2	Mapping of coarse eastward and northward components to the target grid.
Total fitted estimators	29	

Table 1: Learned components in the saved artifact. Deterministic audited policies are reported separately in the text.

3.2 Speed calibration and downscaling

Quantile model-output statistics estimates systematic differences between the supplied predictors and the high-resolution target. The lower, median, and upper speed quantiles are fitted directly, rather than deriving an interval from a point forecast. Coarse eastward and northward components are mapped to the target grid by two compact LightGBM regressors using flow and static-location descriptors [7]. A conformal residual adjustment is then estimated from held historical data. At inference, interval endpoints are sorted and clipped so that $0 \leq q05 \leq q50 \leq q95$.

Treatment is lead specific. One- and seven-day forecasts can use their corresponding supplied forecast leads together with lagged reanalysis and issue-time context. Fourteen days extends beyond the supplied HRES range, so the latest available forecast state is combined with seasonal climatology and historically validated long-range endpoint policies. The resulting endpoint transformations are activated only in calendar-hour and physical cells for which their historical support and validation gates pass.

3.3 Circular direction calibration

Direction is represented through eastward and northward unit-vector components. Residual estimators learn corrections in sine-cosine space, and all blends are performed as vector combinations before conversion back to degrees. This avoids the discontinuity produced by linear operations near north. A shared spatial ridge term captures broad, geographically coherent error, while lead-specific centre and endpoint policies account for differences in forecast uncertainty. Seven-day centre policies are selected by leave-year replay; their exact cross-validation interval-score change is -9.109 . Fourteen-day policies are retained only in issue-hour slots that pass exact 2019–

2020 fine-grid replay.

3.4 Causal external trajectory signal

The sole external forecast resource is GraphCast output from the WeatherBench 2 public archive. The selected 37-level GraphCast model is documented as being trained on ERA5 from 1979 through 2017 [8, 9]. Only causal 00 UTC initializations are used. Eastward and northward wind at 1000 hPa are sampled at +24, +30, +36, and +42 hours for one-day correction and at +168, +174, +180, and +186 hours for seven-day correction.

GraphCast is used as an additional trajectory signal, not as a replacement forecast. It is interpolated to the organizer footprint and blended with the calibrated incumbent. The selected one-day speed branch uses a weight of 0.30 in 12.5% of issue-hour slots; the one-day direction centre uses a 0.30 circular blend; and the seven-day speed branch adjusts centre and interval width in 62.5% of supported cells. No GraphCast model weights or forecast fields are included in the submitted archive.

4 Forecast experiments and results

4.1 Historical promotion evidence

Table 2 reports the principal ablation results used to retain optional components. Reference and candidate values are shown only where they were evaluated on the same branch-specific sample. Changes from different rows are neither directly comparable nor additive because activation masks, folds, and incumbent references differ.

Component	Held periods	Reference	Candidate	Δ score	Activation and strict gate evidence
GraphCast, +1d speed	2018, 2020	–	–	–0.316	12.5% active; both years improve; worst broad-regime change +0.033, below the recorded 0.05 tolerance.
GraphCast, +1d direction	2018, 2020	–	–	–4.857	Year changes –2.330 and –7.385; worst populated broad regime –0.317.
GraphCast, +7d speed	2018, 2020	–	–	–1.996	62.5% active; both years improve; no positive issue block; leave-one-issue-out change –1.850.
+1d context endpoints	2019 / 2020	9.975 / 13.040	9.645 / 11.468	–0.330 / –1.572	61.9–75.0% active; every audited physical regime is non-worse.
+7d fixed endpoints	2016–2020	18.143	17.157	–0.987	37.5% active; 93.95% coverage; every year and affected cell-year is non-worse.
+7d conditional endpoints	2017–2020	–	–	–2.331	Four chronological folds improve; worst populated-regime change is zero.
+14d speed endpoints	2019, 2020	21.943	17.013	–4.931	84.38% active; 91.75% coverage; each year and its worst audited regime improves.

Table 2: Branch-specific historical interval-score evidence. Negative change denotes improvement. These values are retrospective validation diagnostics, not definitive evaluation scores.

Promotion gates also prevent aggregate improvements from concealing localized damage. A stronger one-day GraphCast speed blend, for example, improved its mean score but worsened nested issue-time and fine-grid regimes and was rejected. A separate fine-grid speed-residual branch also failed its transfer gate. Rejected candidates are not stored in the production artifact.

For the selected seven-day GraphCast speed branch, a block bootstrap over issue times produced no non-improving draw in the recorded audit. This is supportive but should not be interpreted as a population-level probability because the branch itself was selected using the same development data.

4.2 Definitive forecast evaluation

The competition scorer matched all 4,196,640 rows to the definitive reference (100% key coverage). Table 3 reports all six raw scores and the platform aggregate. No claim about the unobserved target distribution is inferred from these values.

Interval score	+1 day	+7 days	+14 days
Wind speed	9.511	17.124	16.835
Wind direction	76.400	335.571	315.183
Platform primary score	1.497276		

Table 3: Definitive forecast scores returned by the competition platform. Lower values are better. The primary score is rank based and is not an arithmetic mean of the six entries.

5 Wind-farm siting method

5.1 Centre screening and exact replay

The centre search covers the supplied spatial domain before invoking expensive wake simulation. An initial gross-power screen evaluates 37,835 weather cells and all 23,406 legal centre cells. Historical wind is sampled every sixth day at this stage. Spatially diverse candidates are retained from high-mean, high-worst-year, balanced, near-best, and depth-safe groups. Fourteen centres then receive exact five-year replay at the full available temporal resolution.

A challenger is promoted only if it remains legal, materially improves mean and worst-year capacity factor, avoids annual deterioration, controls maximum wake loss, transfers across the complete 5×5 weather-cell neighborhood, and remains acceptable under the official economic calculation. This screening is intended to identify a robust solution, not to claim a global optimum over continuous latitude and longitude.

5.2 Wake-aware layout optimization

The layout starts from a quality-diversity search, which preserves strong but geometrically different arrangements. The retained geometry is refined by constrained turbine-coordinate moves evaluated against all five historical wind climates. The search performed 340 candidate evaluations and accepted 26 moves. Every accepted move preserves the legal square and minimum spacing.

The hub-height transformation uses the prescribed power-law shear exponent of 0.11 from 125 m to the 170 m IEA 22 MW hub height. Wakes are evaluated using the prescribed Bastankhah-Gaussian PyWake setup [10, 2]. Exact replay after coordinate refinement improves capacity factor in every historical year, raises mean capacity factor by 0.000133, raises its worst-year value by 0.000187, and reduces maximum wake loss by 0.000360.

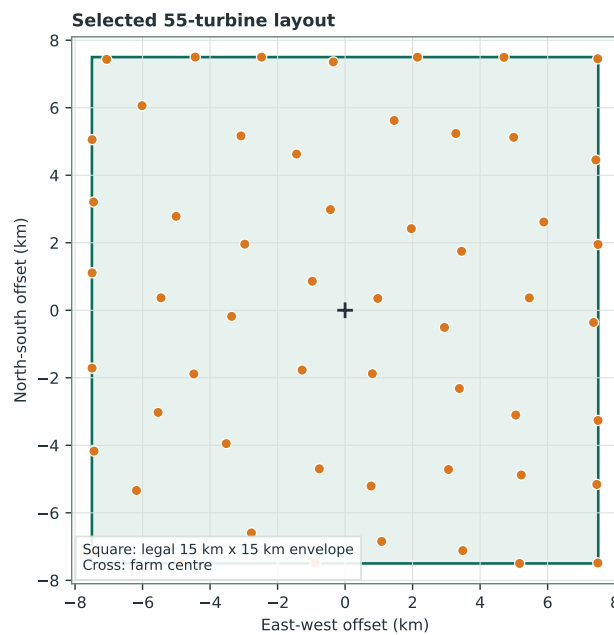


Figure 2: Final turbine coordinates relative to the selected farm centre. The geometry uses the legal envelope while retaining a minimum pairwise separation of 1,554.64 m.

5.3 Farm-power quantile construction

The optional bidding forecast is coupled directly to the selected site and layout. For each of the 96 forecast times, the inference program extracts the nearest 1.3 km grid cell to the farm centre. Let $q_{v,t}^{(r)}$ denote the site wind-speed forecast for $r \in \{0.05, 0.50, 0.95\}$, let $\tilde{d}_t$ be the median direction, and let $\mathcal{L}$ denote the submitted layout. The three energy paths are

$$E_t^{(r)} = 6 \text{ h } P_{\text{farm}} \left[q_{v,t}^{(r)} \left(\frac{170}{125} \right)^{0.11}, \tilde{d}_t, \mathcal{L} \right],$$

where P_{farm} is evaluated with the organizer IEA 22 MW power curve and Bastankhah-Gaussian PyWake model. Thus the conversion includes cut-in, rated-power saturation, and layout-specific wake interaction rather than multiplying wind speed by a constant.

The three transformed paths are sorted at each time to preserve the required lower–median–upper order and clipped to the physical interval $[0, 7,260]$ MWh. This is a compact scenario-propagation approximation: all three

paths share the median direction, so directional uncertainty and the full joint dependence between speed and direction are not integrated. This limitation is preferable to claiming unsupported probabilistic precision, and it avoids fitting an additional power model without an independent power target.

6 Siting results

6.1 Selected design and legal audit

The selected centre is $54.109676^{\circ}\text{N}$, 0.952316°E . Its nearest supplied bathymetry value is 44.14 m and its nearest supplied coastline diagnostic is 69.07 km. Figure 3 places the site within the common high-resolution footprint.

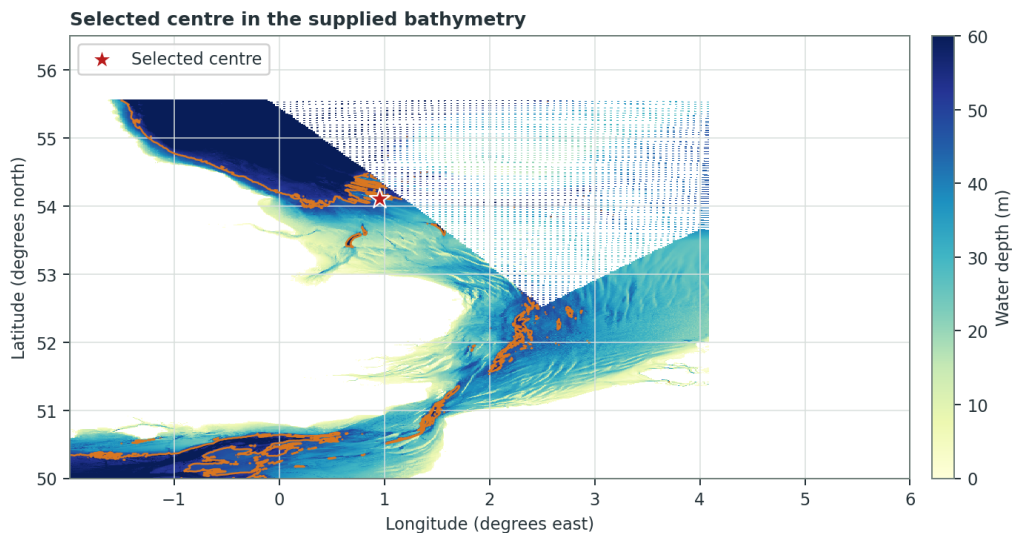


Figure 3: Selected centre on the supplied bathymetry. The orange contour denotes the 50 m fixed-bottom depth limit.

Constraint	Audited value	Requirement	Status
Turbine count and capacity	55; 1,210 MW	exactly 55; 1,210 MW	pass
Centre depth	44.14 m	≤ 50 m	pass
Minimum pair spacing	1,554.64 m	$\geq 1,420$ m	pass
Maximum $ x $	7,495.61 m	$\leq 7,500$ m	pass
Maximum $ y $	7,499.94 m	$\leq 7,500$ m	pass
Zone, sea, and weather footprint	eligible	required	pass

Table 4: Deterministic legal audit of the submitted centre and 55-turbine layout.

6.2 Farm-energy output audit

The inference program constructs the optional bidding fields only after the final gridded forecast has passed its key and quantile checks. Table 5 summarizes the corresponding deterministic audit. This is a format and physics audit rather than a performance score, because no definitive farm-power observations are available locally.

Audit item	Observed value	Status
Time-step coverage	$8 \times 3 \times 4 = 96$	pass
Values per quantile field	96	pass
Finite energy values	288 of 288	pass
Quantile-order violations	0	pass
Energy range	0–7,260 MWh	pass
Source provenance	final wind forecast and trained siting artifact	pass

Table 5: Deterministic audit of the optional farm-energy quantiles written to `submission.json`.

6.3 Like-for-like reference and annual replay

To quantify improvement without using hidden wind, a plain-grid proxy was simulated at 53.5°N , 1.5°E with the same official historical target weather, hub-height transformation, and wake configuration as the selected design. It is a reproducible internal comparator, not the organizers' hidden-wind baseline; the latter cannot be reconstructed from the public data.

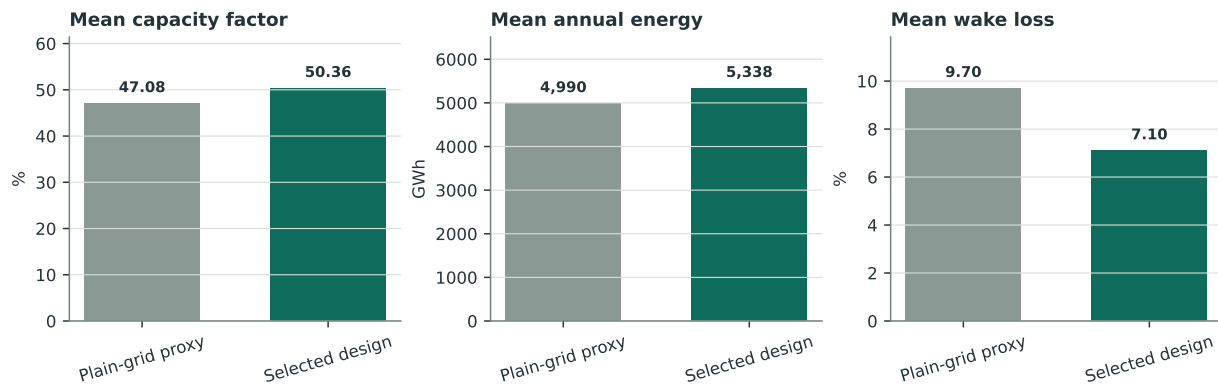


Figure 4: Like-for-like historical comparison between the plain-grid proxy and the selected joint centre-layout design. Capacity factor and annual energy production increase, while wake loss decreases. Both designs are evaluated on the same public historical weather and simulator configuration.

Metric	Plain-grid proxy	Selected design	Difference
Mean capacity factor	47.08%	50.36%	+3.28 percentage points
Worst-year capacity factor	44.29%	48.18%	+3.89 percentage points
Mean AEP	4,989.9 GWh	5,337.7 GWh	+347.7 GWh (+6.97%)
Mean wake loss	9.70%	7.10%	−2.60 percentage points
Maximum annual wake loss	10.27%	7.31%	−2.96 percentage points

Table 6: Public-data reference comparison. Improvements combine centre selection and layout optimization.

Exact annual replay of the selected design is shown in Table 7. The interannual capacity-factor range is 48.18–54.95%, and the annual wake-loss range is 6.93–7.31%.

Weather year	AEP (GWh)	Capacity factor	Wake loss
2016	5,106.9	48.18%	7.31%
2017	5,824.1	54.95%	6.93%
2018	5,279.1	49.80%	7.00%
2019	5,327.4	50.26%	7.31%
2020	5,150.7	48.59%	6.97%
Mean	5,337.7	50.36%	7.10%

Table 7: Exact PyWake replay over the five official historical target years.

6.4 Robustness tests

Three tests probe different forms of siting sensitivity. First, all wind directions are shifted coherently by $-45, -30, -15, 0, +15, +30$, and $+45$ degrees. Capacity factor remains between 50.23% and 50.38%, maximum wake loss is 7.32%, and the coordinate refinement remains beneficial at every shift. Second, 2,000 monthly-block bootstrap resamples preserve within-month temporal dependence. None reverses the refinement improvement; the 1st and 5th percentiles of capacity-factor improvement are 0.000090 and 0.000102. Third, the refined geometry is transferred to 31 nearby legal centres. All 31 pass the recorded every-year mean, worst-year, and wake gates.

These tests do not reveal the hidden wind climate. Their purpose is narrower: to test whether modest, physically interpretable perturbations reverse the historical advantage of the selected geometry.

7 Economics, frugality, and reproducibility

7.1 Economic diagnostics

Under mean historical weather, the official cost calculation gives capital expenditure of EUR 4,478.4 million, annual operating expenditure of EUR 123.1 million, and levelized cost of energy of EUR 88.69/MWh. The latter consists of EUR 65.63/MWh attributed to capital expenditure and EUR 23.06/MWh attributed to operating expenditure. The lowest-production historical year gives EUR 92.70/MWh. These quantities are diagnostics under the prescribed model, not commercial investment forecasts.

7.2 Computational design

The solution uses exactly two source files. `train.py` resolves official and declared external inputs, fits forecast estimators, audits optional branches, performs siting search and replay, evaluates economics, and saves one artifact directory containing both the selected geometry and its farm-power conversion policy. `inference.py` loads those artifacts, processes inference windows sequentially, validates the gridded output, derives the 96 farm-energy triplets, and writes both `predictions.csv` and `submission.json`.

All fitted estimators and numerical libraries are restricted to one processing thread. There is no seed ensemble or model zoo. The external neural weather model is not executed locally; only the required variables, pressure level, issue times, and lead hours are sampled from its public causal trajectory archive. The center search uses inexpensive screening over the full legal domain and reserves exact five-year wake simulation for 14 candidates. The recorded clean training run required 2,107.85 s (35.1 min); elapsed time will vary with hardware and storage.

The complete generation sequence is:

```
python train.py --kit-dir KIT_ROOT --data-root PHASE2_DATA_ROOT \
  --phase1-data-root PHASE1_DATA_ROOT --artifacts-dir artifacts
```

```
python inference.py --kit-dir KIT_ROOT --data-root INFERENCE_DATA_ROOT \
  --artifacts-dir artifacts --output predictions.csv \
  --archive phase2_forecast_submission.zip
```

No previous prediction or submission file is an input. Training creates the learned artifacts, siting design, and recorded power-conversion policy from the declared data; inference uses those artifacts and the official inference predictors. The scored archive places `predictions.csv` and `submission.json` at its root. The latter contains the farm centre, 55 coordinates, turbine key, and three 96-value energy-quantile arrays. Re-running the two entry points therefore regenerates both competition deliverables end to end.

7.3 External resource disclosure

GraphCast is the sole external forecast resource. The 37-level model is documented as trained on ERA5 from 1979 through 2017. Forecast output is obtained from the public WeatherBench 2 archive and used only at causal lead hours. According to the official documentation, the implementation code is Apache-2.0 and the associated materials are CC BY 4.0 [9]. The fields are used as input features and are not redistributed. No third-party operational forecast product is used beyond the HRES inputs supplied with the competition data.

8 Discussion

The historical experiments show that the largest speed improvements arise from lead-specific endpoint correction rather than increasingly complex point models. This is consistent with the interval-score objective: a useful system must correct both conditional centre and uncertainty width. The causal external trajectory signal adds measurable one-day direction and seven-day speed information, but only after masking and conservative blending. Stronger blends were rejected when localized regimes deteriorated.

The definitive results also expose remaining limitations. One-day direction is substantially more accurate than long-range direction, whereas seven- and fourteen-day direction remains difficult. The long-range branches are deliberately conservative because sparse directional regimes can appear stable during development yet fail under a changed atmospheric distribution. Historical gate results should not be read as confidence intervals, and repeated branch selection can make retrospective improvements optimistic.

For siting, the public-data comparison indicates that joint centre and layout optimization can increase historical energy while reducing wakes relative to a plain grid. Nevertheless, the selected centre is represented by the nearest high-resolution weather cell and supplied bathymetry cell, and PyWake remains a model of wake behavior. The hidden wind climate may rank legal centres differently. Annual replay, neighborhood transfer, directional stress, and block bootstrap address complementary sensitivities, but none establishes a global optimum or removes hidden-climate uncertainty.

The farm-energy forecast closes the methodological gap between atmospheric uncertainty and the bidding output, while adding no trained estimator. Its physical bounds and layout consistency are exact by construction. Its probabilistic interpretation is more limited: propagating three marginal speed quantiles through a nonlinear wake model with a common median direction does not recover the full conditional distribution of farm energy. A larger study could estimate the joint speed–direction dependence and calibrate power quantiles directly, but doing so reliably would require independent validation evidence that is unavailable for the definitive period.

Frugality imposes a useful modeling discipline. The system invests computation in exact validation and physically meaningful stress tests rather than a large ensemble of closely related estimators. This improves auditability and permits complete regeneration through two entry points, although it may leave performance available to larger

multiscale neural systems.

9 Conclusion

This work presents a reproducible solution for high-resolution probabilistic wind forecasting, farm-energy estimation, and offshore wind-farm siting. The forecast combines direct quantile calibration, vector-aware circular correction, compact downscaling, causal GraphCast trajectory information, and strict component promotion. The siting solution couples exhaustive legal screening with exact multi-year wake replay and constrained coordinate refinement. A physical wind-to-power stage then generates 96 ordered and capacity-bounded farm-energy triplets from the final forecast and selected layout. All forecast keys, energy-output checks, and siting constraints pass deterministic audits. Historical evidence supports the selected components and layout, while the discussion explicitly separates that evidence from definitive hidden-set performance and from claims of global optimality.

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